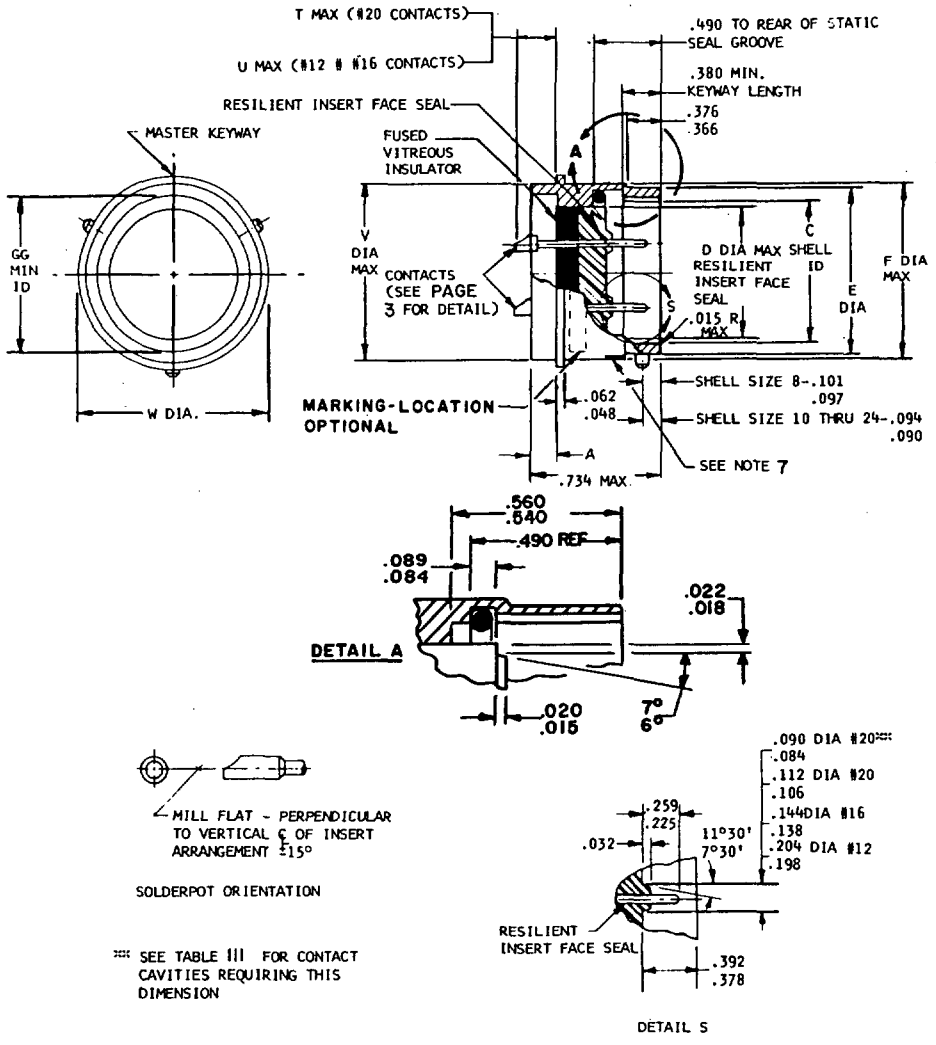
MIL-C-83723/80A
13 NOV 1975
SUPERSEDING
MIL-C-83723/80(USAF)
9 September 1971

MILITARY SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, (CIRCULAR, ENVIRONMENT RESISTING), RECEPTACLE, (SOLDER FLANGE MOUNT, BAYONET COUPLING, SOLDER PIN CONTACT), (SERIES III, CLASSES H AND Y)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the connector described herein shall consist of this document and the latest issue of Specification MIL-C-83723.



FSC 5935

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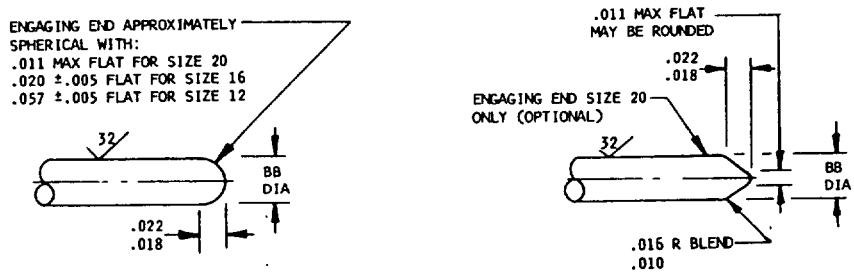
TABLE I. Connector dimensions.

SHELL SIZE	A	C	D	E	F	G	T	U	V	W
8	.108 .078	.428 .433	.312	.536 .531	.561	.091 .097	.330	.400	.500	.760 .720
10	.108 .078	.530 .535	.388	.659 .654	.696	.091 .097	.330	.400	.562	.860 .820
12	.108 .078	.700 .705	.558	.829 .824	.875	.091 .097	.330	.400	.750	1.065 1.025
14	.108 .078	.769 .774	.627	.898 .893	.935	.091 .097	.330	.400	.812	1.110 1.070
16	.108 .078	.896 .901	.772	1.025 1.020	1.062	.122 .128	.330	.400	.937	1.230 1.190
18	.108 .078	1.002 1.007	.860	1.131 1.126	1.187	.122 .128	.330	.400	1.062	1.360 1.320
20	.108 .078	1.127 1.132	.985	1.256 1.251	1.132	.122 .128	.330	.400	1.187	1.450 1.410
22	.108 .078	1.252 1.257	1.110	1.381 1.376	1.437	.122 .128	.330	.400	1.312	1.610 1.570
24	.108 .078	1.377 1.382	1.235	1.506 1.501	1.562	.122 .128	.330	.400	1.437	1.730 1.690

SIZE	Y	Z	FF	GG
8	.474 .484	.620 .602	.505 .515	.406
10	.576 .586	.748 .730	.567 .577	.443
12	.746 .756	.918 .900	.755 .765	.668
14	.816 .826	.988 .970	.817 .827	.668
16	.942 .952	1.114 1.096	.942 .952	.763
18	1.048 1.058	1.220 1.202	1.067 1.077	.862
20	1.172 1.182	1.346 1.328	1.192 1.202	1.108
22	1.298 1.308	1.470 1.452	1.317 1.327	1.204
24	1.422 1.432	1.596 1.578	1.442 1.452	1.388

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ENGAGING END OF CONTACTS



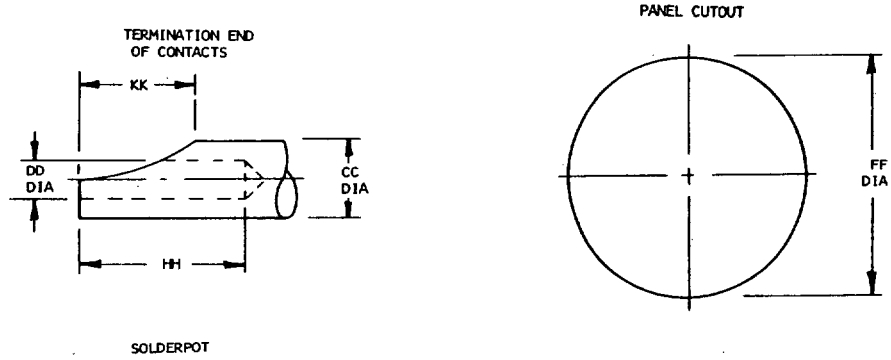


TABLE II. Dimensions of engaging end.

CONT SIZE	BB	CC	DD	HH	KK
20	.041 .039	.088 .061	.048 .042	.188 .109	.110 .068
16	.0635 .0615	.116 .096	.075 .069	.251 .172	.141 .109
12	.095 .093	.151 .131	.122 .112	.251 .172	.141 .109

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INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.000050	.001	.112	2.84	.540	13.72	.860	21.84	1.202	30.53
.003	.08	.116	2.95	.558	14.17	.862	21.89	1.204	30.58
.004	.10	.122	3.10	.560	14.22	.875	22.23	1.220	30.99
.005	.13	.128	3.25	.561	14.25	.893	22.68	1.230	31.24
.010	.25	.131	3.33	.562	14.27	.896	22.76	1.235	31.37
.011	.28	.138	3.51	.567	14.40	.898	22.81	1.251	31.78
.015	.38	.140	3.56	.576	14.63	.900	22.86	1.252	31.80
.016	.41	.141	3.58	.577	14.66	.901	22.89	1.256	31.90
.018	.46	.144	3.66	.586	14.88	.918	23.32	1.257	31.93
.020	.51	.151	3.84	.602	15.29	.935	23.75	1.298	32.97
.022	.56	.172	4.37	.620	15.75	.937	23.80	1.308	33.22
.032	.81	.188	4.78	.627	15.93	.942	23.93	1.312	33.32
.039	.99	.198	5.03	.654	16.61	.952	24.18	1.317	33.45
.041	1.04	.204	5.18	.659	16.74	.970	24.64	1.320	33.53
.042	1.07	.225	5.72	.668	16.97	.985	25.02	1.327	33.71
.048	1.22	.251	6.38	.696	17.68	.988	25.10	1.328	33.73
.057	1.45	.259	6.58	.700	17.78	1.002	25.45	1.346	34.19
.061	1.55	.312	7.92	.705	17.91	1.007	25.58	1.360	34.54
.0615	1.56	.330	8.38	.720	18.29	1.020	25.91	1.376	34.95
.062	1.57	.366	9.30	.730	18.54	1.025	26.04	1.377	34.98
.0635	1.61	.376	9.55	.734	18.64	1.048	26.62	1.381	35.08
.068	1.73	.378	9.60	.746	18.95	1.058	26.87	1.382	35.10
.069	1.75	.380	9.65	.748	19.00	1.062	26.97	1.388	35.26
.075	1.91	.388	9.86	.750	19.05	1.065	27.05	1.410	35.81
.078	1.98	.392	9.96	.755	19.18	1.067	27.10	1.422	36.12
.084	2.13	.400	10.16	.756	19.20	1.070	27.18	1.432	36.37
.088	2.24	.406	10.31	.760	19.30	1.077	27.36	1.437	36.50
.089	2.26	.428	10.87	.763	19.38	1.096	27.84	1.442	36.63
.090	2.29	.433	11.00	.765	19.43	1.108	28.14	1.450	36.83
.091	2.31	.443	11.25	.769	19.53	1.110	28.19	1.452	36.88
.093	2.36	.474	12.04	.772	19.61	1.114	28.30	1.470	37.34
.094	2.39	.484	12.29	.774	19.66	1.126	28.60	1.501	38.13
.095	2.41	.490	12.45	.812	20.62	1.127	28.63	1.506	38.25
.096	2.44	.500	12.70	.816	20.73	1.131	28.73	1.562	39.67
.097	2.46	.505	12.83	.817	20.75	1.132	28.75	1.570	39.89
.101	2.57	.515	13.08	.820	20.83	1.172	29.77	1.578	40.08
.106	2.69	.530	13.46	.824	20.93	1.182	30.02	1.596	40.54
.108	2.74	.531	13.49	.826	20.98	1.187	30.15	1.610	40.89
.109	2.77	.535	13.59	.827	21.01	1.190	30.23	1.690	42.93
.110	2.79	.536	13.61	.829	21.06	1.192	30.28	1.730	43.94

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TABLE III. Insert arrangements requiring reduced diameters for raised seal barriers on contact cavities as indicated.

Shell size	Insert arrangements	Contact cavities
8	-3	1, 2, and 3
10	-6	1, 2, 3, 4, 5, 6

NOTES: (Notes 1 through 6 applicable to contacts only).

1. Contact finish: Gold plate .000050 inch min. In accordance with MIL-G-45204. A suitable underplating shall be used. Silver shall not be used. Finish is applied after sealing.
2. Dimensions are in inches. Dimensions apply after plating.
3. 63/ all over per ANSI B46.1-62 unless otherwise specified.
4. Remove all burrs and sharp edges .003 to .005 R.
5. Diameters to be concentric within .004 TIR.
6. Diameters in the same cross-sectional plane to be concentric with each other within .003 TIR.
7. Polarizing stripe shall be color coded blue.
8. Three (3) pins to be located within .004 either side of (TP) relative to master polarizing keyway.
9. Unless otherwise specified, tolerances are .005 (.13 mm) for three place decimals and .01 (.25 mm) for two place decimals. Angles ±1°.
10. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

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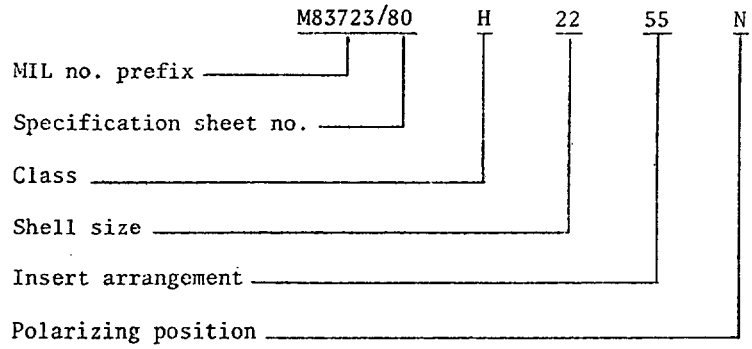
REQUIREMENTS:

Dimensions and configuration: See figure.

This connector mates with MIL-C-83723/75 and 77.

For insert arrangements see MIL-STD-1554.

Part number example:



Custodians:
Army - EL
Navy - AS
Air Force - 85

Review activities:
Army - MI, MU
Navy - EC
Air Force - 11, 15, 17, 80
DSA - ES

User activities:
Army -
Navy -
Air Force -

Preparing activity:
Air Force - 85

(Project 5935-1948-10)

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OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-C-83723/80A Connectors, Electrical, (Circular, Environment Resisting)

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER FSC 5935
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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